

Static Cosmology

Data Exercise 14

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Topic

Automated Catalog Updating

Submission

Submit one ZIP archive containing report.pdf and the requested data table. Record the catalog and calibration versions in the report. The maximum file size is 25 MB.

Problems

1. Solve the first displayed relation used in Automated Catalog Updating. State all assumptions and conventions.

$$r_{cell} = (r_{color}, r_{flux}, r_{phase}, r_{shape}, r_T)$$

2. Estimate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\theta(x) = \theta_0 + \Delta\theta_c G_c(x)$$

3. Analyze the third relation in two limiting regimes and explain the physical meaning of the result.

$$accept = L_{new} \downarrow \wedge L_{atlas} \leq L_{old}$$

Show enough intermediate work that another student can reproduce the calculation.